Factry Historian Module
Design Document

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Introduction

This document is based on five days of research into the Ignition 8.3 Historian API and Factry Historian integration.

Ignition Platform

Ignition is an industrial automation platform for SCADA, IIoT, MES, and more from Inductive Automation. Version 8.3, released in August 2024, is the latest major release. Ignition is written in Java and Kotlin, with version 8.3 requiring Java 17. The platform provides a module SDK that allows users to extend its functionality with custom modules. Gradle is the recommended build tool and package manager.

Ignition documentation is auto-generated from JavaDoc, but Kotlin components are not fully documented. Combined with limited examples for the new 8.3 APIs, this creates challenges for module development. The official Inductive Automation forum provides active support from experts to fill these gaps.

Historian Concept

In industrial automation, a historian is a software component that continuously collects, stores, and serves time-series data from industrial sources (such as PLC tags and devices) for querying, trending, and analysis.

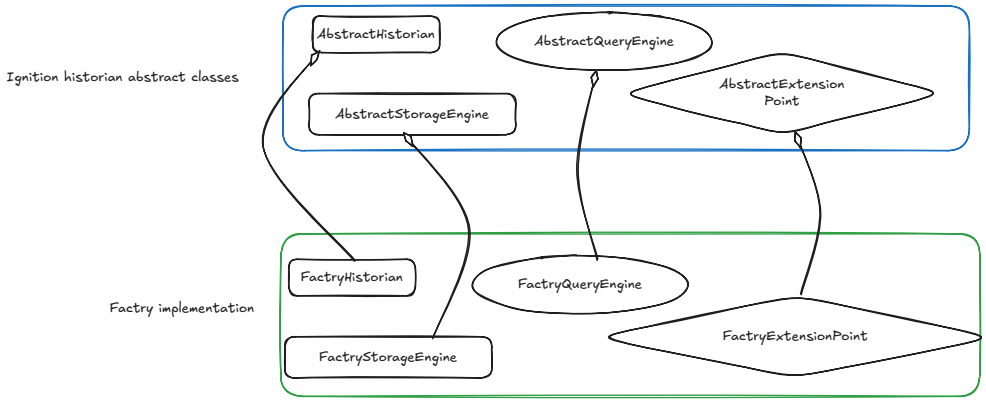
The Ignition module SDK contains the Historian API for third-party historian integration. This allows us to write a custom module for Factry Historian.

Ignition Historian SDK

Version 8.3 introduced a major refactoring of the Historian API. While the core changes are complete, documentation and examples are still limited. The public API is primarily contained in two packages:

- `com.inductiveautomation.historian.gateway.api`
- `com.inductiveautomation.historian.common.model`

Ignition modules use OOP concepts: implementation of a new historian connection is based on inheriting and overriding the abstract classes.



API Structure

To give a quick overview of the new API, here is list of the most important elements.

```
com.inductiveautomation.historian.gateway.api
├── Historian<S>                    - Main historian interface
├── AbstractHistorian<S>            - Base implementation class
├── HistorianManager                - System historian manager
├── config/
│   └── HistorianSettings          - Configuration marker interface
├── query/
│   ├── QueryEngine                - Data retrieval interface
│   ├── AbstractQueryEngine        - Base query implementation
│   ├── browsing/
│   │   └── BrowsePublisher        - Tag browsing API
│   └── processor/
│       ├── RawPointProcessor      - Raw data processing
│       ├── AggregatedPointProcessor - Aggregated data processing
│       └── ComplexPointProcessor  - Complex data processing
├── storage/
│   ├── StorageEngine              - Data storage interface
│   └── AbstractStorageEngine      - Base storage implementation
└── paths/
    └── QualifiedPathAdapter       - Path normalization
```

Ignition Modules Folder Structure

Ignition defines scopes that constrain where parts of a module may execute. We use the GCD scopes, meaning the module's shared code is available in all three scopes:

- G (Gateway): Server-side code running on the Ignition Gateway
- C (Client): Code for Vision Clients and Designer environment
- D (Designer): Designer-only functionality

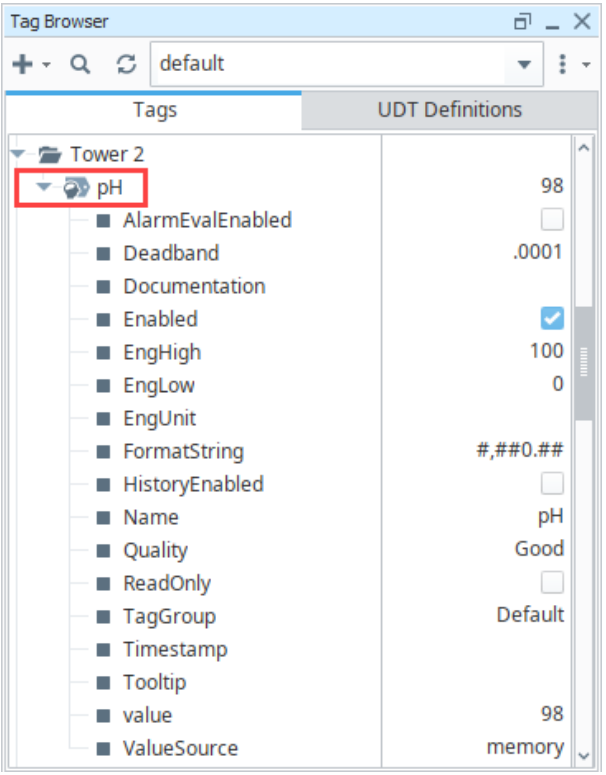
The module folder structure partly reflects Ignition's GCD-scope architecture:

```
factry-historian-module/
├── common/              - Shared code (GCD scope: Gateway, Client, Designer)
│   └── src/           - Module constants and shared interfaces
├── gateway/           - Server-side historian logic (G scope: Gateway)
│   └── src/           - Storage Provider and History Provider implementations
├── client/            - Client runtime code (CD scope: Client, Designer)
│   └── src/           - Vision Client functionality
├── designer/          - Designer-specific code (D scope: Designer)
│   └── src/           - Designer tools and UI components
├── certificates/      - Module signing certificates (*.jks, *.p7b)
├── build/             - Gradle build output
│   └── Factry-Historian.modl  - Final signed module file
├── gradle/            - Gradle wrapper files
└── docs/              - Project documentation
```

Flow of data

Tags are named data points that represent real-time values from industrial sources (PLCs, sensors, OPC servers) or calculated values, serving as the fundamental abstraction for accessing, storing, and scripting against process data throughout the Ignition platform.

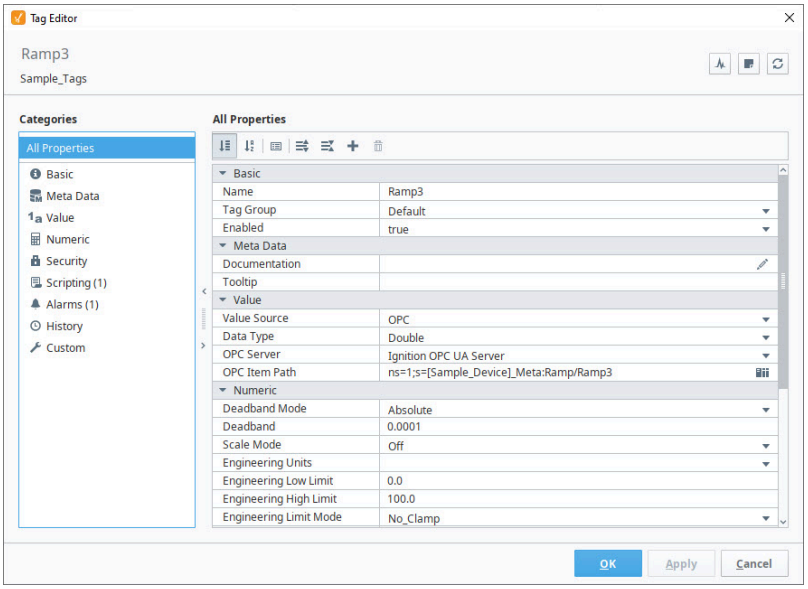
The Tag Browser displays all tags organized by tag provider:



Beyond a simple 'value', each tag has additional properties:

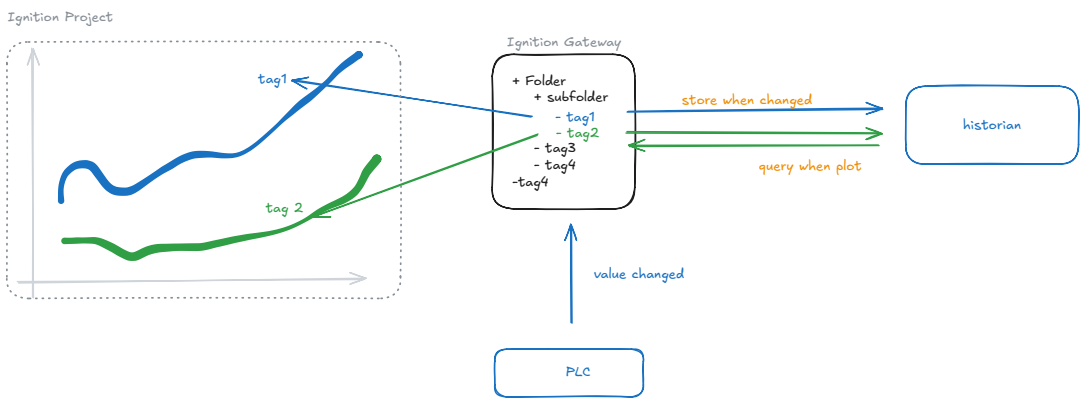
- Metadata: Engineering units, format string, documentation
- Quality: Connection status, staleness indicators
- History: Configuration for historical data storage etc..

The History property is where the Factry Historian connects to a tag. All tag properties can be configured in the Tag Editor, including assigning Factry Historian as the history provider:



Tag Mapping and Data Flow

The following diagram illustrates how tags send and receive historical data:



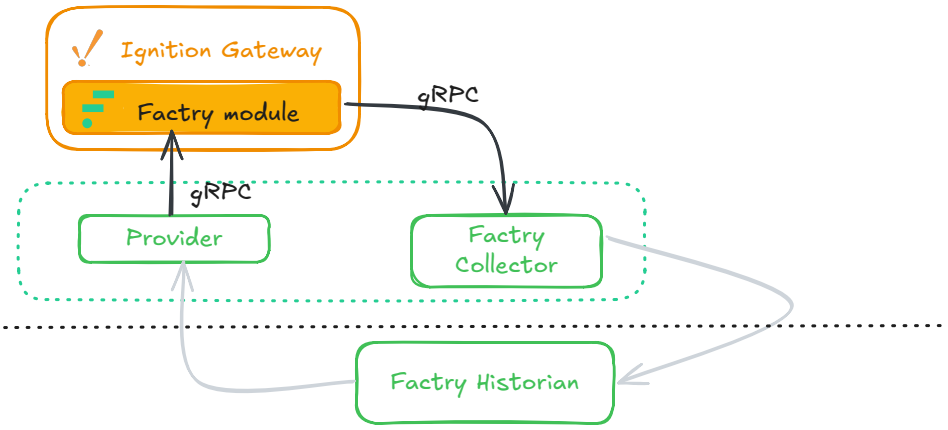
Writing data (Storage): When a tag provider updates a tag value, the Factry module's storage engine is invoked. The implementation should forward the new data point to the Factry Collector (either as a proxy or directly to Factry Historian).

Reading data (Query): When a tag is added to a chart or trend, the Factry module's query engine is invoked. It constructs and sends a query to Factry Historian, then returns the retrieved data for visualization.

Factry Collector Integration

The Factry Collector can serve as an intermediary between Ignition and Factry Historian, providing:

- Data buffering: Store-and-forward capability during network outages
- Compression: Reduce bandwidth usage for high-frequency data
- Local aggregation: Pre-process data before transmission



The module supports two deployment modes:

1. Direct mode: Ignition communicates directly with Factry Historian
2. Collector mode: Ignition sends data through the Factry Collector proxy

Note: The Factry Provider component is not yet implemented, and the Collector may not have all planned features available.

Communication Protocol

The module communicates with Factry services using gRPC, a high-performance RPC framework designed for low-latency, high-throughput communication.

Protocol Buffers (protobuf) define the message formats and service interfaces shared between Factry and the Ignition module. These definitions generate type-safe Java classes for:

- Data point transmission (timestamps, values, quality codes)
- Query requests and responses
- Configuration and metadata exchange

This approach ensures consistent serialization, backward compatibility, and efficient binary encoding for industrial data volumes.

Ignition Python Functions

With proper implementation of the historian module, the historian functionality is available from Ignition's Python scripting environment. This chapter documents the available functions.

Supported Functions

The Factry Historian module implements 6 of the available historian functions: Functions marked as out of scope are not implemented in this version.

browse
Returns a list of browse results for the specified Historian.

deleteAnnotations
Deletes desired annotations from the specified Historian.

queryAggregatedPoints
Queries aggregated data points for the specified historian.

queryAnnotations
Queries annotations for the specified Historian.

queryMetadata
Queries metadata for the specified Historian.

queryRawPoints
Queries raw data points for the specified historian.

storeAnnotations
Store a list of annotations to the specified Historian.

storeDataPoints
Store a list of data points to the specified Historian.

storeMetadata
Store a list of metadata to the specified Historian.

updateRegisteredNodePath
Updates the existing historical path for a stored historian node to the n...

1. browse

Returns a list of browse results for the specified historian.

Syntax 1:

```
system.historian.browse(rootPath, browseFilter)
```

Parameters:

- `rootPath` (String): The root path to start browsing from.
- `browseFilter` (Dictionary[String, Any]): A dictionary of browse filter keys. Keys are listed below.

Filter Keys:

- `dataType` : Represents the data type on the tag. Valid values can be found on the Tag Properties page.
- `valueSource` : Represents how the node derives its value. Generally only used by nodes with a tag type of "AtomicTag".
- `tagType` : The type of the node (tag, folder, UDT instance, etc). A list of possible types can be found on the Tag Properties page.
- `typeId` : Represents the UDT type of the node. If the node is a UDT definition, then the value will be None. If the node is not a UDT, then this filter choice will not remove the element. Functions best when paired with a tagType filter with a value of UdtInstance.
- `quality` : Represents the "Bad" and "Good" quality on the node. All other quality codes are ignored.
- `maxResults` : Limits the amount of results that will be returned by the function.

Syntax 2:

```
system.historian.browse(rootPath[, snapshotTime][, nameFilters][, maxSize][, recursive][,
continuationPoint][, includeMetadata])
```

Parameters:

- `rootPath` (String): The root path to start browsing from.
- `snapshotTime` (Date): The snapshot time to browse at. [optional]
- `nameFilters` (List): A list of name filters to apply to the browse results. [optional]
- `maxSize` (Integer): The maximum number of results to return. [optional]
- `recursive` (Boolean): Whether to browse recursively. Accepted values are True and False. False is the default, meaning that the browse will only return data directly inside the root path. [optional]
- `continuationPoint` (String): The continuation point to continue browsing from. [optional]
- `includeMetadata` (Boolean): Whether to include metadata in the browse results. [optional]

2. queryAggregatedPoints

Queries aggregated data points for the specified historian.

```
system.historian.queryAggregatedPoints(paths, startTime, endTime[, aggregates][, fillModes][,
columnNames][, returnFormat][, returnSize][, includeBounds][, excludeObservations])
```

Parameters:

- `paths` (List): A list of historical paths to query aggregated data points for.
- `startTime` (Date): A start time to query aggregated data points for.
- `endTime` (Date): An end time to query aggregated data points for.
- `aggregates` (List): A list of aggregate functions to apply to the query. [optional]
- `fillModes` (List): A list of fill modes to apply to the query. [optional]
- `columnNames` (List): A list of alias column names for the returned dataset. [optional]
- `returnFormat` (String): The desired return format for the query. [optional]
- `returnSize` (Integer): The maximum number of results to return. [optional]

- `includeBounds` (Boolean): Whether to include the bounds in the query results. [optional]
- `excludeObservations` (Boolean): Whether to exclude observed aggregated data points in the query results. [optional]

3. queryRawPoints

Queries raw data points for the specified historian.

```
system.historian.queryRawPoints(paths, startTime, endTime[, columnNames][, returnFormat][,
returnSize], includeBounds[, excludeObservations])
```

Parameters:

- `paths` (List): A list of historical paths to query raw data points for.
- `startTime` (Date): A start time to query raw data points for.
- `endTime` (Date): An end time to query raw data points for.
- `columnNames` (List): A list of alias column names for the returned dataset. [optional]
- `returnFormat` (String): The desired return format for the query. [optional]
- `returnSize` (Integer): The maximum number of results to return. [optional]
- `includeBounds` (Boolean): Whether to include the bounds in the query results.
- `excludeObservations` (Boolean): Whether to exclude observed raw data points in the query results. [optional]

Example:

```
# Query a historical tag path and display raw data from the past minute
end = system.date.now()
start = system.date.addMinutes(end, -1)

myDataset = system.historian.queryRawPoints(
    ["[default]_Simulator_/Random/RandomInteger1"],
    start,
    end,
    includeBounds=False
)

for row in myDataset:
    print row[0], row[1]
```

4. queryMetadata

Queries metadata for the specified historian.

```
system.historian.queryMetadata(paths[, startDate][, endDate])
```

Parameters:

- `paths` (String): A list of historical paths to query metadata for.
- `startDate` (Date): A start time to query metadata for. This parameter is optional, unless an end time is specified.
- `endDate` (Date): An end time to query metadata for. If specifying an end time, a start time must be provided. [optional]

5. storeMetadata

Stores metadata for the specified historian.

Syntax 1:

```
system.historian.storeMetadata(metadata)
```

Parameters:

- `metadata` (List): A list of metadata.

Syntax 2:

```
system.historian.storeMetadata(paths, timestamps, properties)
```

Parameters:

- `paths` (List): A list of historical paths.
- `timestamps` (List): A list of timestamps.
- `properties` (Dictionary): A dictionary of desired properties to be stored as historical metadata.

6. storeDataPoints

Stores data points to the specified historian.

Syntax 1:

```
system.historian.storeDataPoints(datapoints)
```

Parameters:

- `datapoints` (List): A list of data points.

Syntax 2:

```
system.historian.storeDataPoints(paths, values[, timestamps][, qualities])
```

Parameters:

- `paths` (List): A list of historical paths.
- `values` (List): A list of historical values.
- `timestamps` (List): A list of timestamps. [optional]
- `qualities` (List): A list of qualities. [optional]

Example using DataPoint:

```
from com.inductiveautomation.historian.common.model import DataPoint

datapoint = DataPoint(
    "histprov:test:/sys:myGateway:/prov:default:/tag:me",
    42,
    system.date.now(),
    192
)

system.historian.storeDataPoints([datapoint])
```

Summary of Required Features

The Factry Historian module consists of two main components:

Collector (Storage Provider)

Responsible for writing data from Ignition to Factry Historian:

- Store tag data points with timestamps and quality codes
- Store tag metadata changes
- Support direct mode (Ignition → Factry Historian)
- Support collector mode (Ignition → Factry Collector → Factry Historian)
- Handle connection failures and reconnection

Provider (History Provider)

Responsible for reading historical data from Factry Historian:

- Query raw data points for specified time ranges

- Query aggregated data points with configurable aggregation functions
- Query tag metadata
- Browse available historical tags
- Support pagination for large result sets

In addition to these, we must ensure that the following components also function correctly.

Configuration

- Gateway configuration page for connection settings
- Support for multiple Factry Historian instances
- Tag history provider selection in Tag Editor

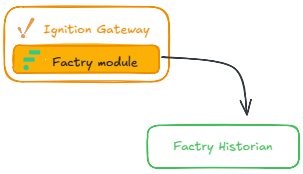
Python scripts

- the scoped 6 python methods

Appendix

Milestones

1. Demo
   Factry module sends the new datapoints to the Factry Historian directly.



1. Historian Collector Full implementation of the Factry module's collector part as described here.
2. Historian Provider Full implementation of the Factry module's provider part as described here.

Links

Ingition 8.3 documentation

https://www.docs.inductiveautomation.com/docs/8.3/intro

System historian functions

https://docs.inductiveautomation.com/docs/8.3/appendix/scripting-functions/system-historian

Custom historian forum

https://forum.inductiveautomation.com/t/ignition-8-3-building-a-custom-tag-historian-module/100725